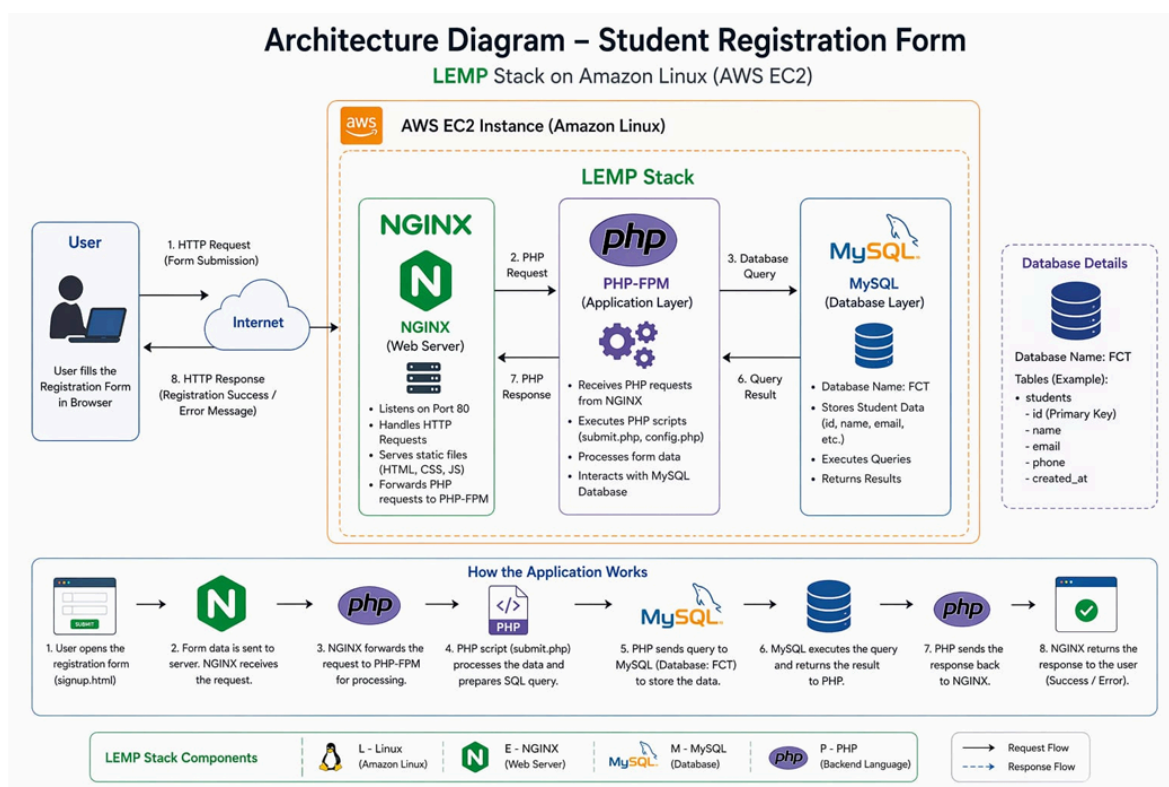


Registration Form Deployment (Dynamic Website LEMP Stack)

1)Introduction:

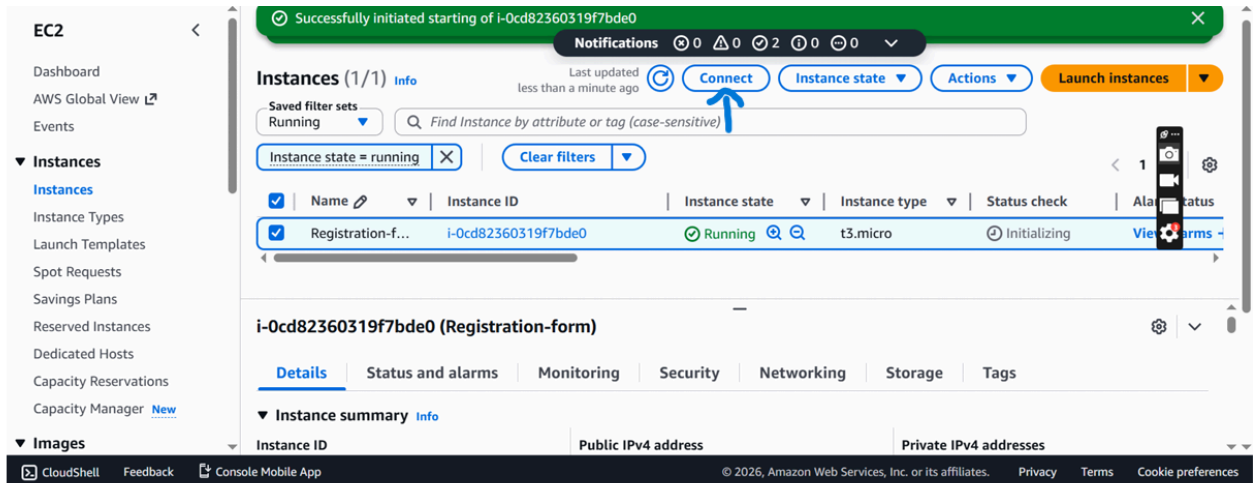
This is my first dynamic website called Student Registration Form. In this Website where users can enter their details like name, email, and other information. The website is built using: Frontend → HTML , CSS Backend → PHP Database → MySQL It is deployed on an Amazon Linux EC2 server using the LEMP stack (Linux, NGINX, MySQL, PHP). This setup helps the website to run online and store student data securely.

2) Architecture Diagram:



3)Deployment :

1:create EC2 instance



2:Install LEMP : `sudo yum update sudo yum install nginx php php-fpm php-mysqlnd mariadb-server -y`

3:Start Services `sudo systemctl start nginx sudo systemctl enable nginx sudo systemctl start php-fpm sudo systemctl enable php-fpm sudo systemctl start mariadb sudo systemctl enable mariadb`

```
ec2-user@ip-172-31-34-205:~$
CGroup: /system.slice/nginx.service
└─26228 "nginx: master process /usr/sbin/nginx"
   └─26229 "nginx: worker process"
      └─26230 "nginx: worker process"

Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Starting nginx.service: nginx - web server.
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal nginx[26226]: nginx: the config file /etc/nginx/nginx.conf
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal nginx[26226]: nginx: configuration file /etc/nginx/nginx.conf
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Started nginx.service: nginx - web server.

[ec2-user@ip-172-31-34-205 ~]$ sudo systemctl status php-fpm
● php-fpm.service - The PHP FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-04-22 14:04:27 UTC; 9min ago
     Main PID: 26225 (php-fpm)
    Status: "Processes active: 0, idle: 5, Requests: 0, slow: 0, Traffic: 0.00req/sec"
       Tasks: 6 (limit: 1014)
      Memory: 11.2M
         CPU: 97ms
    CGroup: /system.slice/php-fpm.service
            └─26225 "php-fpm: master process (/etc/php-fpm.conf)"
               └─26231 "php-fpm: pool www"
                  └─26232 "php-fpm: pool www"
                     └─26233 "php-fpm: pool www"
                        └─26234 "php-fpm: pool www"
                           └─26235 "php-fpm: pool www"

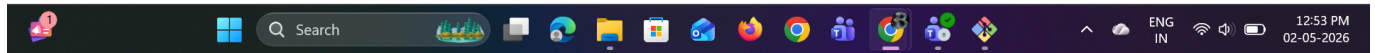
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Starting php-fpm.service: The PHP FastCGI Process Manager.
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Started php-fpm.service: The PHP FastCGI Process Manager.

lines 1-18/18 (END)...skipping...
● php-fpm.service - The PHP FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-04-22 14:04:27 UTC; 9min ago
     Main PID: 26225 (php-fpm)
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            └─26225 "php-fpm: master process (/etc/php-fpm.conf)"
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                     └─26233 "php-fpm: pool www"
                        └─26234 "php-fpm: pool www"
                           └─26235 "php-fpm: pool www"

Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Starting php-fpm.service - The PHP FastCGI Process Manager.
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Started php-fpm.service - The PHP FastCGI Process Manager.
```

4:check status

5:hit public ip we can see LEMP page

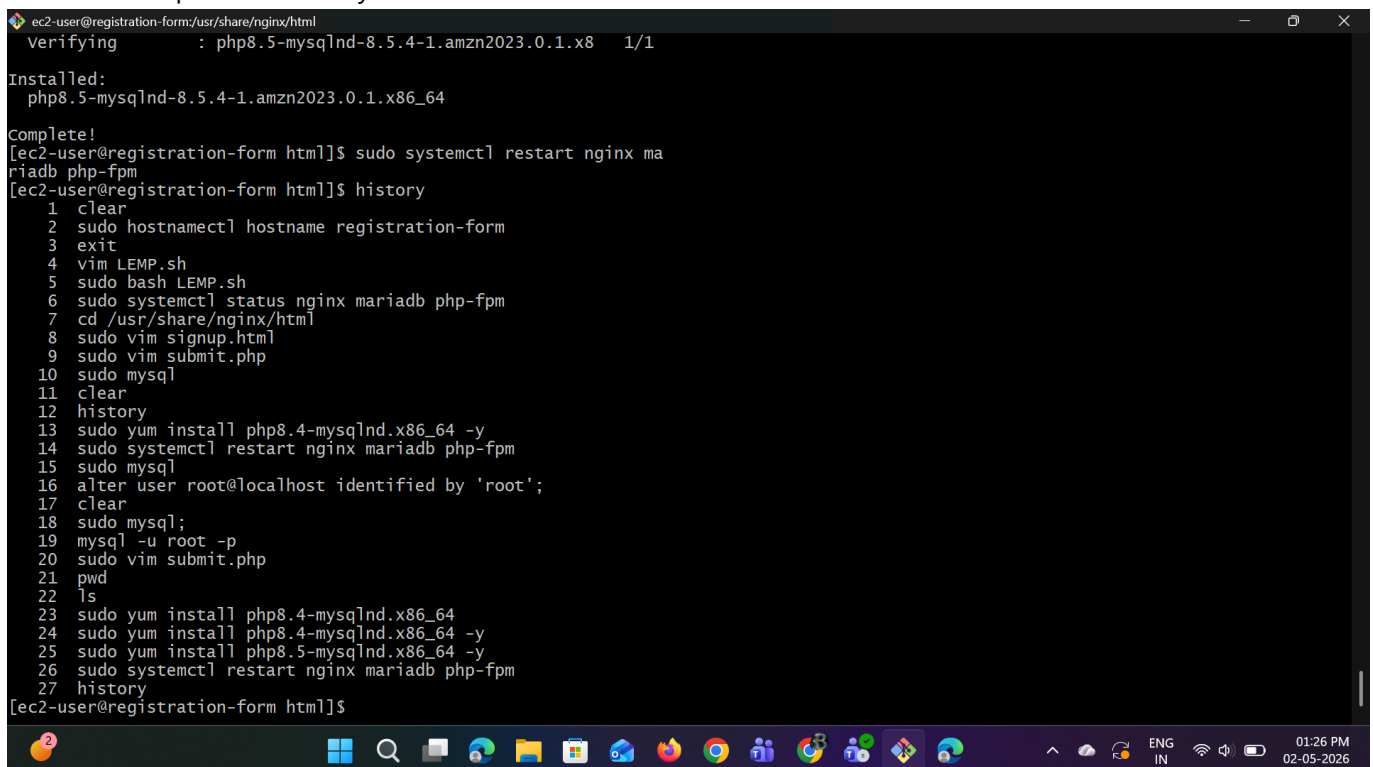


6:Setup Database Login: mysql -u root -p Create database: CREATE DATABASE FCT; Create table: USE FCT; CREATE TABLE students (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100),email VARCHAR(50));

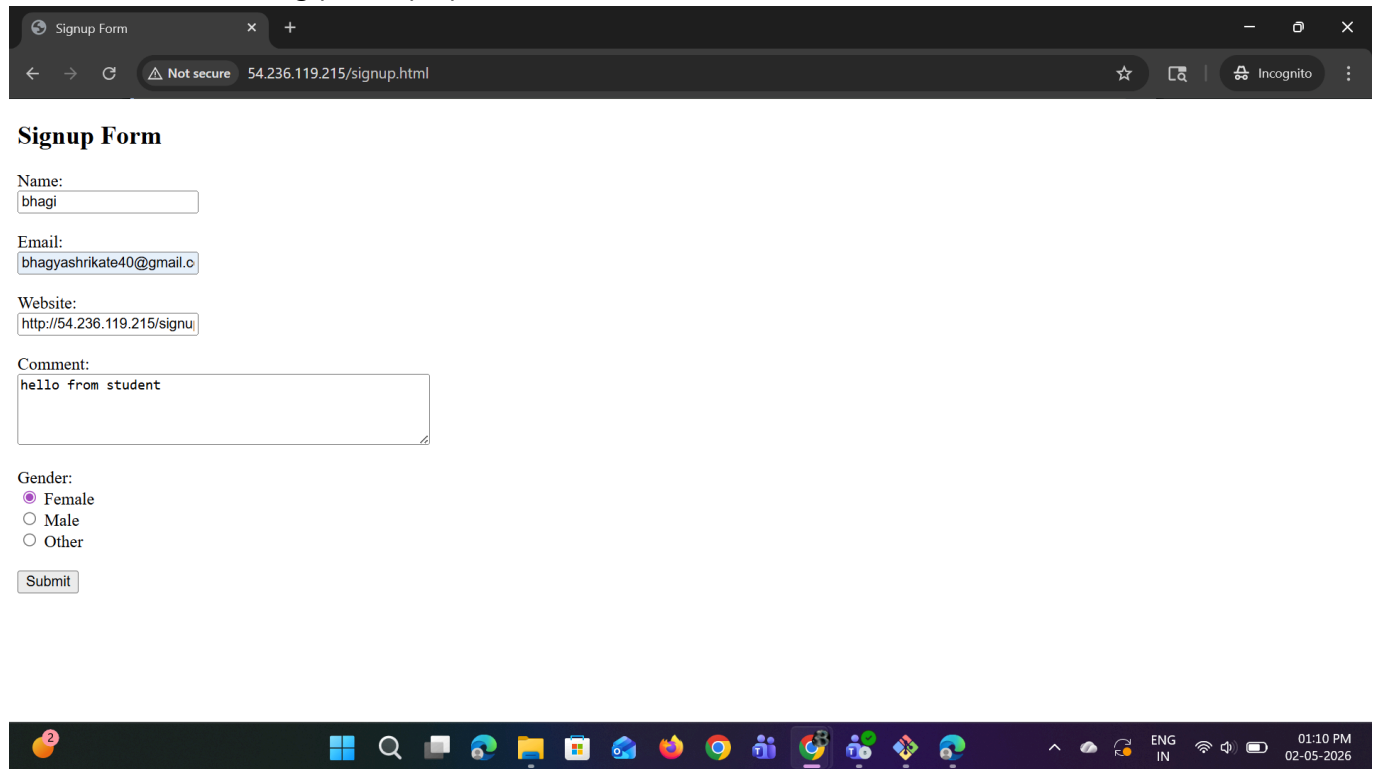
7: Create files signup.html - for ui submit.php -for backend

8: Download connector sudo yum install php8.5-mysqlnd.x86_64 -y

9: After all steps restart the system



10: Access website using public ip open browser



Signup Form

Name:
bhagi

Email:
bhagyashrikate40@gmail.com

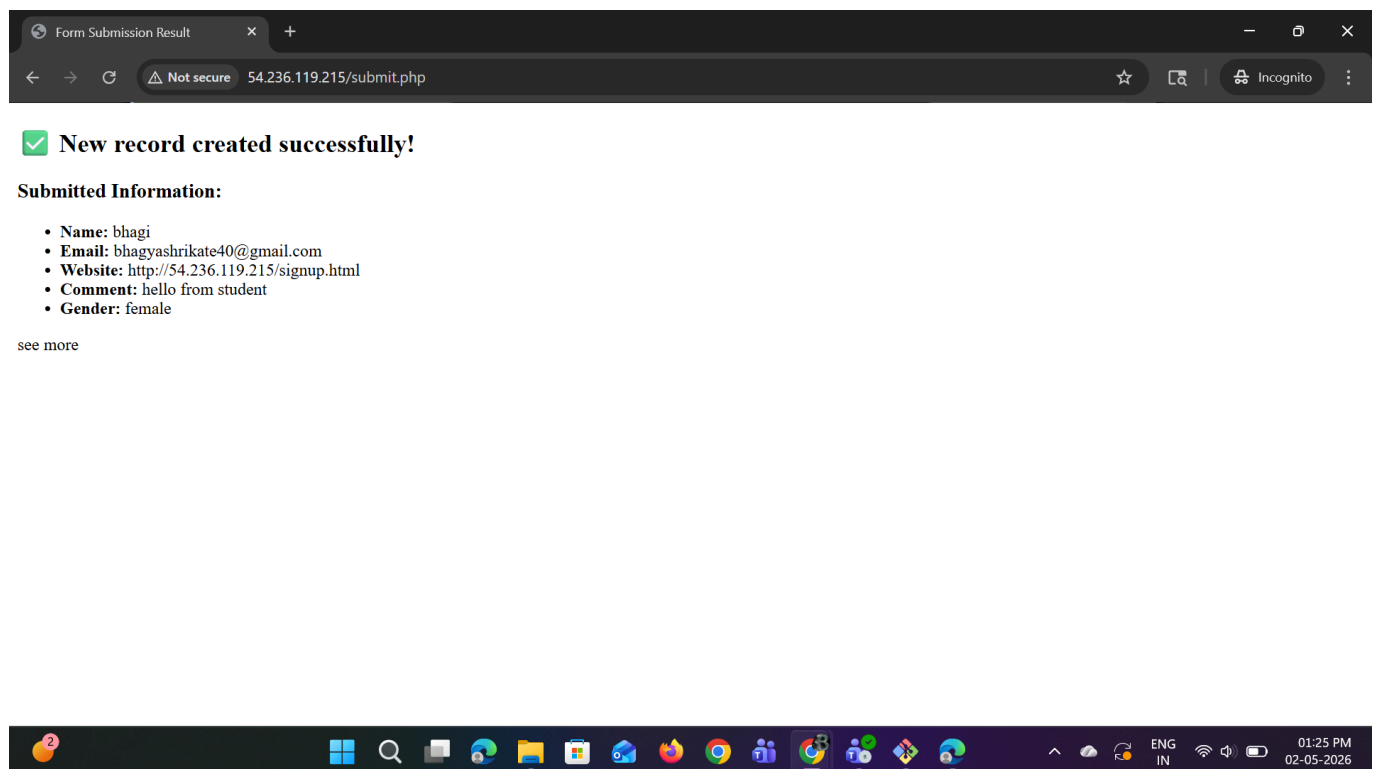
Website:
http://54.236.119.215/signup

Comment:
hello from student

Gender:
☒ Female
☐ Male
☐ Other

Submit

11: Fill form and submit data



Form Submission Result

Not secure 54.236.119.215/submit.php

✓ New record created successfully!

Submitted Information:

- Name: bhagi
- Email: bhagyashrikate40@gmail.com
- Website: http://54.236.119.215/signup.html
- Comment: hello from student
- Gender: female

see more

4) Summery

This project is a dynamic student registration form built using the LEMP stack. It shows how PHP and MySQL work together to collect and store user data. The website handles user input, processes it on the server, and saves it in the database. This is an important step in my journey towards DevOps, and cloud computing.